Mohammad Alaei

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Education

- **Master of Science** — University of Tabriz, Iran
  Computer Engineering – Computer Systems Architecture; ***GPA: 18.93/20*** — *Sep 2018 – Sep 2021*
  Thesis: Deep Learning–Based Non-Contact Reconstruction of Frontal EEG from Facial Expressions
  Supervisor: Dr. Hadi S. Aghdasi

- **Bachelor of Science** — University of Tabriz, Iran
  Computer Engineering – Computer Hardware Engineering; ***GPA: 17.11/20*** — *Sep 2014 – Jul 2018*
  Final Project: Designing and Building a Robot to Play the Bell Musical Instrument
  Supervisor: Dr. Hadi S. Aghdasi

Research Interests

- **Personalization and Recommender Systems**
- **Human–AI Interaction and Cognitive Computing**
- **Computational Modeling of Human Behavior**
- **Affective Computing**

Academic Experiences

- **Research Internship** — Remote
  National Elite Foundation of Iran and JobVision (Part-time) — *Oct 2020 - March 2021*
  - Identified critical employer-demanded soft skills through research and analysis
  - Designed quantification methods to evaluate these skills via online cognitive games
  - Evaluated reliability and efficacy of selected games and documented key assessment tools
  - Synthesized findings into actionable recommendations informing game-based hiring assessments
  - Contributed to a 6-member team evaluating cognitive games to assess soft skills

- **Software Developer (VR Developer)** — University of Tabriz
  National Elite Foundation of Iran (Part-time) — *Oct 2019 - Sep 2020*
  - Developed a VR educational game for 3D geometry, deployed in multiple schools and recognized for enhancing comprehension of complex STEM concepts
  - Implemented immersive VR functionalities using C#, Unity, and HTC Vive to boost 3D geometry learning
  - Collaborated in a 10-member multidisciplinary team, contributing to core VR system design and implementation
  - Designed intuitive menus and interfaces to optimize usability and student learning experience

- **Research Assistant** — University of Tabriz
  Humanoid Robots and Cognitive Technology Research Lab (Part-time) — *Sep 2018 - Feb 2019*
  - Reviewed assistive technology literature for cognitive and functional impairments to guide future research
  - Synthesized and presented research findings at weekly lab meetings, informing discussions and project planning
  - Mentored students on projects and theses, providing guidance on experimental design and methodology
  - Managed lab resources to ensure availability of tools and smooth execution of experiments

Manuscripts & Preprints

- **Mohammad Alaei**, Shabnam Oskouei, Hadi S. Aghdasi: "Deep Learning–Based Non-Contact Reconstruction of Frontal EEG from Facial Expressions", Submitted to *Physica Scripta*, 2025.
- Seyedeh Mahrokh Alaei, **Mohammad Alaei**, Arman Zafaranchi: "An Explainable Human Facial Expression Recognition Model Based on CNN, Gabor Filters, and SVM", Under Review at *The Visual Computer*, 2025.

Academic Achievements

- Member of the **Exceptional Talent Group**, University of Tabriz, Iran — *2018-2021*
- Achieved the **First Rank** Among Students in My Master's Degree, University of Tabriz, Iran — *2021*
- Admitted as an **Exceptional Talent** for a Master's Degree, University of Tabriz, Iran — *2018*
- Achieved the **Third Rank** Among Students in My Bachelor's Degree, University of Tabriz, Iran — *2018*

Professional Experiences

- **Founder & Research Engineer**
  Kavida *Mar 2025 – Present*
  - Designed the scalable architecture and backend for a goal-oriented networking platform
  - Defined feature representations, similarity metrics, and data pipelines supporting collaborator-matching research.
  - Built a full-stack system enabling social-graph expansion and profile-level analytics.

  Technologies: Python, Django, PostgreSQL, Telegram Bot API, React.js

- **Contract Software Engineer**
  EFT Project (via Herrmann Innovations) *Sep 2023 – May 2025*
  - Engineered a real-time vehicle suspension analysis platform with IoT and computer vision, achieving 90% diagnostic reliability
  - Developed Raspberry Pi WiFi camera streaming system, boosting 4K FPS by 50% and reducing setup complexity by 40%
  - Managed and optimized DevOps infrastructure, improving data reliability to 95% and security by 80%

  Technologies: Python, Django REST API, React.js, JavaScript, HTML/CSS, PostgreSQL, Nginx, Raspberry Pi, MJPG Streamer, Git, DevOps

- **Research Software Engineer & Full-Stack Developer**
  Freelance *Mar 2018 - Present*
  - Designed and deployed full-stack web applications, real-time experimental platforms, and automation tools, delivering scalable, multilingual, and research-grade systems
  - Engineered algorithms and prototypes for robotics, cryptography, and cognitive neuroscience with multithreading, sensor integration, and precise real-time data acquisition
  - Collaborated in multidisciplinary teams to integrate UI/UX, database modeling, and hardware-software interfacing, ensuring data integrity, reproducibility, and robust system performance

  Technologies: Python, Flask, Django, C#, JavaScript, SQL, multithreading, real-time systems, robotics frameworks

Certificates

- Supervised Machine Learning: Regression and Classification, DeepLearning.AI and Stanford University *Nov 2024*
- The Interactive Track and the Course Project of Deep Learning, Neuromatch Academy *Jul 2023*
- Game Development Workshop in VR/AR, University of Tabriz *Nov 2019*

Skills

- **Languages**: Python, Java, C/C++, C#, SQL, JavaScript
- **Frameworks**: Pytorch, TensorFlow, Keras, Scikit-Learn, OpenCV, NumPy, Matplotlib, Pandas, Django, Flask, FastAPI, React, Bootstrap
- **Tools**: Unity, Codevision AVR, Git, Proteus, MySQL, SQLite, PostgreSQL
- **Platforms**: Linux, Web, Windows, Raspberry Pi, Arduino

Language Skills

- **English**: Academic Proficiency
- **Azerbaijani**: Mother Tongue
- **Persian**: Native

References

- **Hadi S. Aghdasi, Full Professor, Department of Electrical and Computer Engineering, University of Tabriz, Iran**: Email: aghdasi@tabrizu.ac.ir, aghdasi.ha@gmail.com, Phone: +98 41 3339 3753, Mobile: +98 914 406 0184
- **Leyli Mohammad Khanli, Full Professor, Department of Electrical and Computer Engineering, University of Tabriz, Iran**: Email: l-khanli@tabrizu.ac.ir, Phone: +98 41 3339 3725, Mobile: +98 914 313 8652
- **Mina Zolfy Lighvan, Associate Professor, Department of Electrical and Computer Engineering, University of Tabriz, Iran**: Email: mzolfy@tabrizu.ac.ir, Phone: +98 41 3339 3759, Mobile: +98 914 415 9281